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Q1. Harshad number

ANSWER

```
n=int(input("Enter a number:"))
temp=n
sum=0
while temp>0:
      digit=temp%10
       sum=sum+digit
       temp=temp//10
if n % sum==0:
     print("Harshad Number")
else:
    print("Not Harshad number")
print("Harshad number from 1 to 500:")
for i in range(1,501):
      temp=i
       sum=0
while temp>0:
          digit=temp%10
           sum=sum+digit
            temp=temp//10
if i % sum==0:
  print(i ,end=" " )
```

Q2. GCD

CODE

```
a,b=map(int,input().split())
while b≠0:
    temp=a
    a=b
    b=temp%b
print("GCD is",a)
```

INPUT

```
6 3
```

OUTPUT

```
(no output)
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst,target):
seen=set()
pairs=[ ]
for num in lst:
     complement=target-num
     if complement in seen:
         pairs.append((complement,num))
     seen.add(num)
  return pairs
data=[int(x) for x in input("Enter numbers:").split()]
t=int(input("Target: "))
result=find_pairs(data,t)
print("pairs: ",result)
```


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Q4. Bitwise Operators

ANSWER

```
a=0b1101-binary-1101-13
b=0o25-octal-25-21
c=0x1A-hexadecimal-26
a&b(and)
a=13=01101
b=21-10101
13&21=00101-5
result
5=00101
26-11010
5|26=11111=31

bin(31)='0b11111'
oct(31)='0o37'
hex(31)='0x1f'
result>>2=31>>2=7
result<<1=31<<1=62
~result=~31=-32
DATA TYPES
a:int
b:int
c:int
result:int
```

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low≤high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]≤arr[mid]:
            if arr[low]≤target<arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if arr[mid]<target≤arr[high]:
                low=mid+1
            else:
                high=mid-1
    return -1
data=[4,5,6,7,0,1,2]
t=int(input("Target: "))
result=rotated_binary_search(data,t)
print("Found at: ",result)
```

INPUT

```
6
```

OUTPUT

```
Target: Found at:  2
```